

Wind-OI: A Real-Time CFD-Driven Olfactory Interface with Wind Stimulus for Virtual Dynamic Odor Environments

Shuo Li*
School of Art Design and Media
East China University Of Science and
Technology
Shang Hai, China
shuolils@ecust.edu.cn

Masafumi Uda
School of Engineering
Institute of Science Tokyo
Tokyo, Japan
uda.m.aa@m.titech.ac.jp

Takamichi Nakamoto
Institute of Integrated Research
Institute of Science Tokyo
Yokohama, Japan
nakamoto.t.ab@m.titech.ac.jp

Abstract

While visual and auditory feedback in Virtual Reality are highly developed, realistic olfactory interaction remains limited. Traditional isotropic diffusion models and pre-computed Computational Fluid Dynamics (CFD) ignore real-time hydrodynamic perturbations caused by the user's body acting as a physical obstacle and thermal source. To address spatiotemporal challenges in odor localization within weak wind environments, we propose Wind-OI (Olfactory Interface with Wind-Stimulus). The system solves the Navier-Stokes equations using GPU-accelerated Real-time CFD, Wind-OI dynamically couples user motion and thermal plumes to synchronize nasal concentration fields with facial airflow. Balancing physical ecological validity with perceptible cues, we established an anthropomorphic modeling strategy with wind stimulus compensation. Sensory tests calibrated a 0.2 m/s wind perception threshold. Multimodal evaluations demonstrated that wind tactile cues improved direction discrimination accuracy from 31.25% to 93.75%. Ultimately, Wind-OI contributes a novel HCI design principle: utilizing low-latency wind cues to mitigate multisensory conflict and enhance spatial presence.

CCS Concepts

• **Human-centered computing** → **Virtual reality**; • **Computing methodologies** → *Real-time simulation*.

Keywords

Virtual olfaction, Real-time computational fluid dynamics, Olfactory interface, Multimodal interaction, Spatiotemporal interaction

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*Co-corresponding author: Shuo Li.



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1 Introduction

In the context of Virtual Reality (VR) successfully reproducing visual, auditory, and tactile sensations, olfactory feedback has garnered significant attention due to its profound impact on memory and cognition [1, 3]. There are two main pathways to realizing olfactory VR: the synthesis of scent components and the construction of olfaction-related spatiotemporal interaction. However, existing isotropic diffusion models assume a windless environment, while pre-computed CFD methods presuppose user paths; both result in significant spatiotemporal misalignment between the virtual odor distribution and the user's current dynamic physical state (e.g., walking, body heat convection) [2, 4, 6]. The absence of dynamic flow field information not only diminishes immersion but also complicates the identification of direction in odor source localization tasks. The small concentration gradients and weak ambient wind further obscure the necessary directional cues for precise localization [7]. To address this, we constructed the Wind-OI system. Our core contributions include: (1) Utilizing GPU acceleration technology to calculate the physical obstruction and thermal perturbation of odor diffusion by the user's body in real-time; (2) Proposing a design strategy based on physical model comparison, introducing non-contact wind tactile sensations to compensate for the limitations of high-fidelity anthropomorphic models in single-modality olfactory direction discrimination. We highlight HCI opportunities where synchronized odor-wind cues support attention guidance and wayfinding, enhance spatial presence, and mitigate multisensory conflict when olfactory cues are weak or delayed.

2 System Implementation

2.1 Real-time CFD Framework

To resolve the conflict between computational cost and real-time interaction, this study adopts a GPU parallel computing architecture derived from prior research [5]. While Lagrangian particle methods are common, they struggle with computational bottlenecks when simulating millions of molecules in enclosed spaces with complex thermal convection. Therefore, we adopted the Lattice Boltzmann Method (LBM). This approach maintains stable computation times regardless of molecule density, making it exceptionally robust for scaling to larger virtual rooms while ensuring real-time interactivity. The solver is encapsulated as a Dynamic Link Library (DLL) embedded in Unity, achieving millisecond-level synchronization (update occurs about every 1.25 ms) between the fluid field and user position (Fig. 1). In addition, the VR program was created with Unity 2022.3.20f1 and the CFD program was written in CUDA.

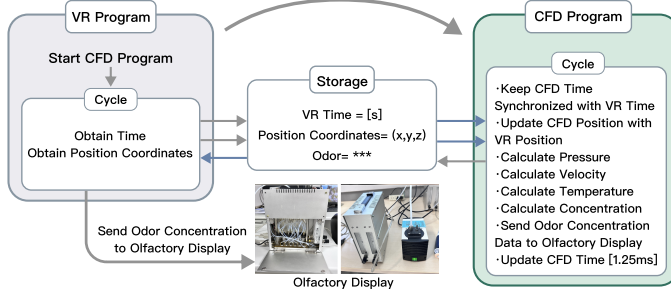


Figure 1: Real-time computational fluid dynamics system calculation process.

The hardware setup utilizes an Intel i7-13700KF processor and an NVIDIA RTX4090 graphics card.

The physics engine is based on the weakly compressible Navier-Stokes equations [8], solved together with the pressure development, temperature transport, and concentration transport equations. The transport model of odorant molecules, based on the diffusion characteristics of volatile substances, enables dynamic interaction with user perception via the virtual olfactory display system, providing physical grounding for spatial presence.

To restore authentic physical interaction, the system introduces key thermodynamic boundary settings: The body surface temperature is set to 305 K (32°C) and the ambient temperature to 293 K (20°C) to simulate the real human thermal plume. The odor source flow velocity is set to 0.2 m/s (initial concentration 0.0667), and the walls in the virtual space for sensory testing are configured with a flow velocity of 0 m/s, ensuring that the velocity at the wall grids remains unchanged.

2.2 HumanModel Selection & Design Rationale

Within the virtual environment, the human body functions not merely as a passive observer but as a dynamic obstacle that fundamentally alters the topological structure of the flow field. To determine the optimal simulation model, we compared a simplified cubic model (HumanModel 1 in Fig. 2a) and an anthropomorphic model (HumanModel 2 in Fig. 2b) within a simulation room (3 m × 3 m × 2.7 m). The domain was configured with a ventilation window (0.4 m × 0.4 m), two identical obstacle boxes (0.4 m × 0.4 m × 1.1 m), and an odor source positioned 1.1 m above the boxes (Fig. 2c). Based on pre-experimental optimizations balancing computational efficiency and spatial precision, we adopted a CFD grid resolution of 5 cm cubic cells, resulting in a total lattice count of 194,401. This high-resolution, millisecond-level mapping ensures that as users naturally turn their heads to 'sniff' the environment, the minute variations in facial airflow and odor concentration are rendered synchronously, significantly reducing motion-to-olfaction latency and enhancing spatial presence. To quantify this interaction, the odor concentration perceived by the user was calculated as the average concentration within a cubic volume positioned 0.15 m in front of the user's head.

Simulation results revealed that HumanModel 1 forces airflow to bypass the body due to leg closure, creating a pronounced physical obstruction. This results in a facial front-to-back odor concentration

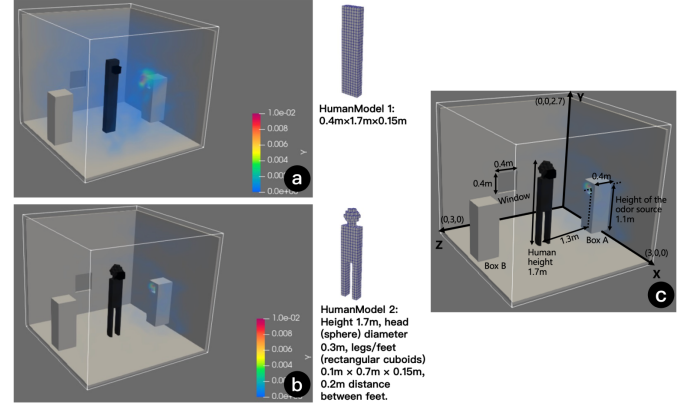


Figure 2: CFD Simulation results of spatial odor diffusion for two models under identical parameter. (a) rectangular, (b) model with leg gap, (c) two same boxes, window, and odor source placement.

difference of up to 12.6 times (Front: 4.51×10^{-4} , Back: 3.58×10^{-5}), providing strong single-modality olfactory directional cues. Conversely, although HumanModel 2 aligns more closely with real physical laws (high ecological validity), it introduces a perceptual challenge. Its complex geometry and leg gaps allow airflow to pass through, diminishing the physical obstruction effect seen in HumanModel 1, which drastically reduces the concentration difference to 2.2 times (Front: 2.73×10^{-4} , Back: 1.24×10^{-4}). To balance this physical realism with interactive usability, we selected HumanModel 2 as the core model and strategically introduced facial wind stimulus (tactile sensations). By utilizing the wind speed difference between the two sides of the face, the system compensates for the insufficiency of olfactory directional cues in this weak fluid environment.

2.3 Determination of Key CFD Simulation Parameters

To determine the initialization time required for the odor field to reach a quasi-steady state, we analyzed the temporal evolution of the concentration distribution within a simulation domain configured with identical dimensions and obstacle settings as described in Section 2.2. The system was observed to stabilize at approximately $t=180$ s. Fig. 3 presents the volumetric data at this timestamp under varying visualization thresholds (referred to as "Set Range" in ParaView). This parameter acts as a scalar cut-off, rendering only gas concentrations within the specified interval to isolate regions of interest. Decreasing the lower bound (from Fig. 3a to 3d) progressively reveals fainter peripheral gas distributions. While Fig. 3a highlights the dense core driven by the thermal plume, Fig. 3d exposes the full diffusion envelope. The results confirm that by $t=180$ s, thermal convection has established a robust transport pathway to the nasal region regardless of the visualization sensitivity. Consequently, this duration was justified as the pre-computation phase for the real-time simulation framework.

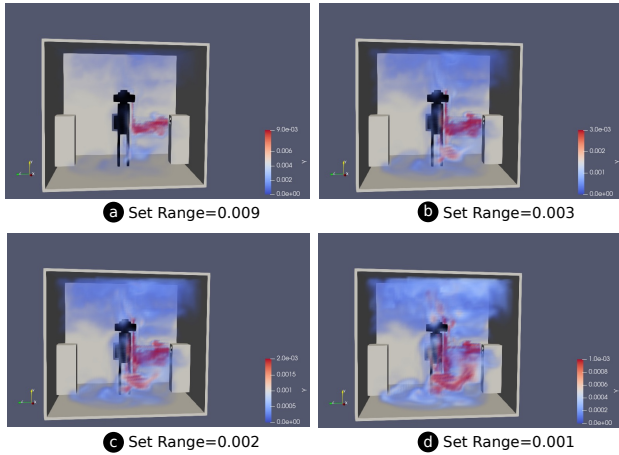


Figure 3: Visualization of CFD simulation data at $t=180s$. Sub-images (a)–(d) display the odor plume morphology with decreasing concentration display thresholds. Lowering the threshold (from 0.009 to 0.001) reveals the spatial continuity of low-density gas, confirming effective coverage of the facial region via thermal convection.

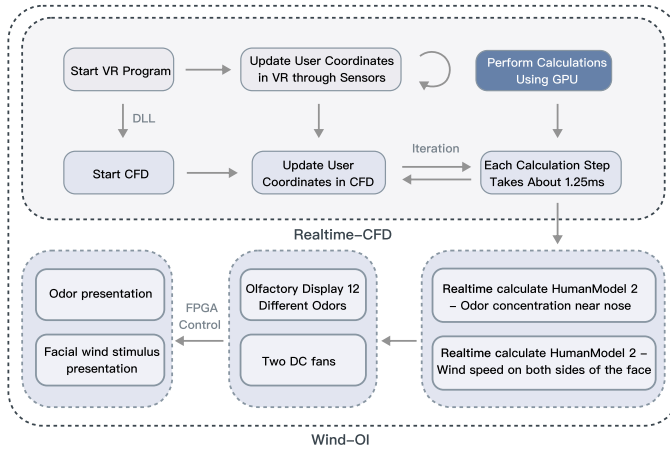


Figure 4: Technical pipeline for odor and wind presentation in the Wind-OI system.

2.4 Hardware Architecture and Data Mapping

The hardware integrates a head-mounted display (Meta Quest 3), a 12-channel solenoid valve olfactory display (Ono Denki, Japan), and two direct current fan (DC fans). A Field-Programmable Gate Array (FPGA) development board receives CFD calculation data to form closed-loop feedback (Fig. 4): (1) Olfactory Mapping: Nasal concentration (approx. 10^{-5} magnitude) is linearly mapped to a 0-100% duty cycle of PWM signal to drive the solenoid valves. (2) Tactile Mapping: The flow velocity difference on both sides of the face is normalized to drive the fans.

3 Sensory Testing

To ensure system parameters align with human perceptual characteristics, we conducted calibration tests on two levels. We ensured that all participants had no olfactory impairments or allergies to the essential oils used. The cohort was selected to represent the core VR/AR user demographic, encompassing diverse professional backgrounds such as engineering, design, and computer science and technology. The experimental protocol was approved by the Institutional Review Board (IRB) of the authors' home institution, and all participants provided written informed consent prior to the study. Before the formal sessions, we provided detailed briefings and standardized guidance on experimental procedures and safety precautions to ensure data consistency and participant well-being.

3.1 Tactile Threshold

We employed a DP70 hybrid anemometer to measure wind speed at a distance of 3.5 cm from four fan models (see Fig. 5a) using a double-blind protocol. Five subjects (3 males, 2 females, age 27.6 ± 3.9) completed four sets of continuous trials across six incremental voltage levels. The results demonstrated a significant positive correlation between wind speed and subjective perception ratings across all models ($R^2=0.89-0.93$, see Fig. 5(b-e)). Using a 5-point Likert scale (1=None, 5=Strong), subjects could distinctly perceive wind sensation and the clarity of wind direction (Rating ≥ 3) when the wind speed reached or exceeded 0.2 m/s. Consequently, we established 0.2 m/s as the perception threshold for non-contact tactile (wind) feedback. Furthermore, DC Fan 4 was selected as the wind generation device for Wind-OI. Due to its higher driving voltage and larger dimensions, DC Fan 4 exhibited the steepest psychophysical response curve, providing high-intensity perceptual characteristics that are well-aligned with the interaction requirements for odor source localization in weak wind environments.

3.2 Multimodal Efficacy

We recruited sixteen healthy participants (8 males, 8 females; age: 28.6 ± 3.8) to evaluate the design strategy and its impact on spatial orientation perception. To isolate the variables, we neutralized auditory interference and positioned subjects 0.5 m from the stimulus source. We then assessed directional discrimination (N, S, E, W) across three conditions: (1) olfactory-only, using a 2:1 mixture of cis-3-hexen-1-ol and ethanol; (2) tactile-only, with an 0.8 m/s airflow; and (3) a synergistic olfactory-tactile condition. Results (Table 1) show that olfactory cues alone yielded only 31.25% accuracy, whereas tactile cues significantly boosted performance to 87.50%. The multimodal combination reached 93.75% accuracy, highlighting how tactile airflow effectively compensates for olfactory limitations and sharpens directional perception in dynamic environments. Future studies will explore sub-threshold wind speeds (<0.2 m/s) where tactile cues are subliminal, positioning the odor gradient as the primary conscious driver for navigation.

4 Discussion & Conclusion

Wind-OI solves the pain points of distortion and latency in traditional static models by coupling human thermal effects with motion parameters in real-time. From an HCI perspective, Wind-OI suggests a multisensory interaction pattern: pairing olfaction with

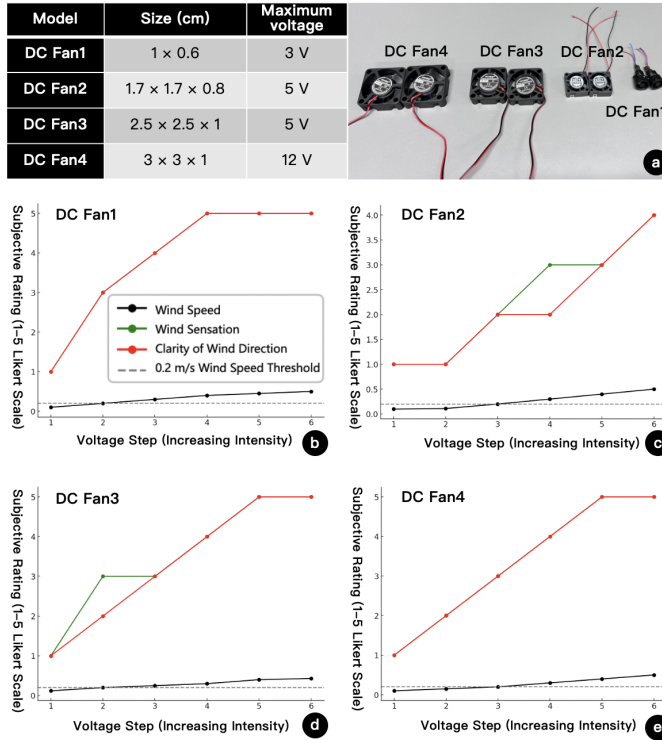


Figure 5: (a) Parameters of DC fan models. (b-e) Psychophysical test results showing the relationship between physical wind and subjective perception. The X-axis represents six incremental voltage steps (e.g., 0.7V to 3V for DC Fan 1, it does not start when the voltage is less than 0.7V), and the Y-axis represents subjective ratings on a 5-point Likert scale. The gray dashed line indicates the determined 0.2 m/s wind speed threshold.

Table 1: Experimental Results of Multimodal Direction Perception

Test Condition	Odor	Wind	Multimodal Synergy (Odor+Wind)
Direction Discrimination Accuracy	31.25%	87.50%	93.75%

congruent, low-latency wind cues to guide attention and wayfinding, strengthen spatial presence, and reduce multisensory conflict under spatiotemporal uncertainty.

This paper constructs the first facial fluid olfactory system integrating Real-time CFD. Through physical model comparison, we established an anthropomorphic modeling strategy fused with wind stimulus compensation and calibrated key perceptual parameters, providing an experimental platform with physical credibility and multimodal compensatory mechanisms for future multisensory VR applications. Ultimately, Wind-OI contributes a multisensory design principle: pairing olfaction with congruent, low-latency wind

cues to disambiguate odor direction, enhance spatial presence, and mitigate multisensory conflict in dynamic VR interactions.

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